GRNScope Consensus Analysis Proposal

Replacing 30 repeated runs with single-run Borda-style consensus metrics

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Recommendation: run each selected algorithm once, then combine ranked edge lists using normalized Borda-style consensus evidence. Direction and sign should be calculated separately using only algorithms that can vote on those properties.

Executive Summary

The current pipeline estimates confidence by running each algorithm 30 times. This is scientifically useful, but it makes analysis very slow. A faster and more explainable approach is to run each algorithm once and then aggregate algorithm rankings across methods.

This proposal treats GRNScope as a consensus-analysis platform. Each algorithm contributes evidence through its ranked edge list. Directed methods vote on direction, signed methods vote on activation versus repression, and methods that cannot provide those properties abstain from those votes.

Current 30-run approach	Proposed one-run consensus approach
Measures repeat stability within each algorithm.	Measures agreement across algorithms using one run per algorithm.
Scientifically useful, but very slow for an interactive web app.	Much faster and better aligned with a consensus-analysis platform.
Uses terms like confidence and stability from repeated runs.	Uses clearer labels: consensus evidence, method support, direction/sign agreement.

Why Raw Algorithm Weights Should Not Be Averaged

Different algorithms output edge weights on different scales. A larger raw value from one method is not necessarily stronger than a smaller value from another method.

```
GENIE3 weight:   0.004
PIDC weight:     12.7
Pearson weight:  0.82
SCODE weight:   -3.4
```

The safer method is to use each algorithm's raw weights only to rank edges within that algorithm, then convert ranks into normalized evidence scores before combining algorithms.

Proposed Metrics

Metric	Question Answered	How It Is Computed
Consensus evidence	Is this regulation strongly supported across algorithms?	Normalized Borda-style rank evidence averaged across selected algorithms.
Method support	How many algorithms independently support this edge?	Count algorithms whose normalized edge evidence passes a support threshold.
Direction agreement	Do directed methods agree on A -> B versus B -> A?	Evidence-weighted vote from directed algorithms only; undirected methods abstain.
Direction coverage	How much evidence can actually vote on direction?	Directed evidence divided by total edge evidence.
Sign agreement	Do signed methods agree on activation versus repression?	Evidence-weighted vote from signed algorithms only; unsigned methods abstain.
Sign coverage	How much evidence can actually vote on sign?	Signed evidence divided by total edge evidence.

1. Consensus Evidence

Consensus evidence measures how strongly a regulation is supported across all selected algorithms. It is conceptually close to BEELINE's BORDA aggregation.

```
b_a(e) = 1 - (rank_a(e) - 1) / (N_a - 1)

ConsensusEvidence(e) = weighted average of b_a(e)
```

Here, e is one edge, a is one algorithm, rank_a(e) is the edge rank from that algorithm, and N_a is the number of ranked edges. A top-ranked edge is near 1.0, a middle-ranked edge is near 0.5, and a weak or missing edge is near 0.

```
Example evidence:
GENIE3:     0.95
GRNBoost2:  0.90
PIDC:       0.80
Pearson:    0.20

ConsensusEvidence = 0.7125
```

2. Method Support

Method support counts how many algorithms independently support an edge above a chosen threshold, such as top 10 percent or normalized evidence >= 0.9.

```
Support(e) = number of algorithms where b_a(e) >= threshold
```

3. Direction Agreement and Coverage

Direction agreement asks whether directed algorithms agree on A -> B versus B -> A. Undirected algorithms still contribute to consensus evidence, but they abstain from direction voting.

```
DirectionVote = sum over directed algorithms of weight * (forward evidence - reverse evidence)

DirectionAgreement = abs(DirectionVote) / total directed evidence
DirectionCoverage = directed evidence / total evidence
```

Coverage is important. A high direction agreement with low direction coverage means the directed algorithms agree, but most total evidence came from undirected methods.

Example

For the pair FOXL2 and NR5A1, suppose three directed algorithms and one undirected algorithm report the following evidence:

```
GENIE3:     FOXL2 -> NR5A1, evidence 0.90
GRNBoost2:  FOXL2 -> NR5A1, evidence 0.80
SCRIBE:     NR5A1 -> FOXL2, evidence 0.40
PIDC:       FOXL2 -- NR5A1, evidence 0.95  (undirected, so no direction vote)

DirectionVote = +0.90 +0.80 -0.40 = 1.30
Total directed evidence = 0.90 +0.80 +0.40 = 2.10
DirectionAgreement = 1.30 / 2.10 = 61.9%
```

Interpretation: the directed algorithms lean toward FOXL2 -> NR5A1, but one directed method supports the opposite direction. PIDC still strengthens the evidence that the two genes are related, but it does not vote on arrow direction.

4. Sign Agreement and Coverage

Sign agreement asks whether signed algorithms agree that an edge is activating or repressing. Unsigned algorithms can support the edge, but they abstain from sign voting.

```
Activation = +1
Repression = -1
Unknown / unsigned = 0

SignVote = sum over signed algorithms of weight * evidence * sign

SignAgreement = abs(SignVote) / total signed evidence
SignCoverage = signed evidence / total evidence
```

For example, if SCODE and SINCERITIES support activation but PPCOR supports repression, sign agreement reflects the evidence-weighted balance between those votes.

Example

For the edge FOXL2 -> NR5A1, suppose the signed and unsigned methods report:

```
SCODE:        activating, evidence 0.80
SINCERITIES:  activating, evidence 0.70
PPCOR:        repressing,  evidence 0.30
GENIE3:       unsigned,    evidence 0.95  (no sign vote)

SignVote = +0.80 +0.70 -0.30 = 1.20
Total signed evidence = 0.80 +0.70 +0.30 = 1.80
SignAgreement = 1.20 / 1.80 = 66.7%

Total evidence = 0.80 +0.70 +0.30 +0.95 = 2.75
SignCoverage = 1.80 / 2.75 = 65.5%
```

Interpretation: signed algorithms lean toward activation, but there is some disagreement. GENIE3 supports the edge strongly, but because it is unsigned, it cannot tell whether the regulation is activating or repressing.

Recommended User-Facing Labels

If GRNScope uses one run per algorithm, it is better to avoid calling the main score confidence. With one run, the platform measures cross-method consensus rather than repeat stability.

- Consensus evidence
- Method support
- Direction agreement
- Direction coverage
- Sign agreement
- Sign coverage

> **Plain interpretation: consensus evidence asks whether the edge is likely supported; direction agreement asks which way the arrow points; sign agreement asks activation versus repression; coverage shows how much evidence was eligible to vote.**

Implementation Recommendation

- Run each selected algorithm once.
- Convert each algorithm's ranked edge list into normalized Borda-style evidence.
- Use normalized BORDA as the primary consensus evidence score.
- Compute method support from the number of algorithms passing the support threshold.
- Compute direction agreement using only directed algorithms.
- Compute sign agreement using only signed algorithms.
- Display direction and sign coverage to prevent overinterpretation.

References

- Pratapa et al. Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. Nature Methods, 2020.
- BEELINE documentation: BLEvaluator.py Borda aggregation output.
- Marbach et al. Wisdom of crowds for robust gene network inference. Nature Methods, 2012.